

A Comprehensive Domain Generalization Framework for Prostate MRI Segmentation with Patch Mixing and Risk Extrapolation — Supplementary Materials

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A. Automated Post-processing with Level-set method

To reduce the dependence of the level-set post-processing on manually selected parameters, we further explored an unsupervised domain-policy search strategy. The raw network prediction was first treated as an identity candidate, and several conservative level-set configurations were used as additional candidate policies. Let I denote the target image and M_0 denote the raw network prediction. For each candidate policy θ , we obtain a refined mask $M_\theta = F_\theta(I, M_0)$ and select the policy by minimizing an unsupervised score:

$$\mathcal{S}(M_\theta) = E_{\text{region}}(I, M_\theta) - \lambda_b E_{\text{boundary}}(I, M_\theta) + \lambda_c R_{\text{change}}(M_\theta, M_0) + \lambda_t R_{\text{topology}}(M_\theta).$$

Here, E_{region} measures the Chan–Vese region consistency inside and outside the mask, E_{boundary} measures the average image-gradient magnitude along the predicted boundary, R_{change} penalizes excessive area or pixel-wise changes from the raw prediction, and R_{topology} penalizes fragmented masks with multiple large connected components. Candidates with excessive area change, excessive XOR difference from M_0 , or multiple large components are rejected. The final policy is selected at the target-domain level by averaging $\mathcal{S}(M_\theta)$ over all target cases:

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \frac{1}{N_D} \sum_{i=1}^{N_D} \mathcal{S}(M_\theta^{(i)}).$$

To avoid unnecessary post-processing, the identity prediction is kept unless a level-set candidate gives a sufficiently lower unsupervised score. It should be noted that target annotations were not used for policy or parameter selection, but only for reporting the final evaluation metrics. The automatic policy search achieved an average DSC of 90.32%, which is comparable to the result obtained using grid-searched parameters. This suggests that the level-set post-processing can be potentially automated without manually tuning the parameters for each individual case.

B. Additional visual examples of Domain Patch Mixing.

To further illustrate the anatomical validity and qualitative behavior of the proposed Domain Patch Mixing (DPM), we

provide additional visual examples of the generated image-mask pairs in this supplementary material. Due to page limitation, only one representative example is included in the main manuscript. The supplementary examples cover different source-domain combinations and different prostate appearances. It can be observed that DPM consistently performs foreground-constrained patch mixing and preserves most of the prostate structure in the mixed image-mask pairs. Compared with other augmentation strategies, the generated samples by DPM exhibit better local structural continuity and introduce less severe anatomical disruption around the target region. In particular, although the mixed patches contain intensity and texture variations from different domains, the resulting masks remain spatially coherent and do not show obvious fragmented or anatomically implausible boundaries. These additional visualizations, together with the quantitative comparison in Table V, further support that DPM can generate effective intermediate-domain perturbations while maintaining no obvious severe anatomical discontinuity is observed in the displayed examples and boundary quality.

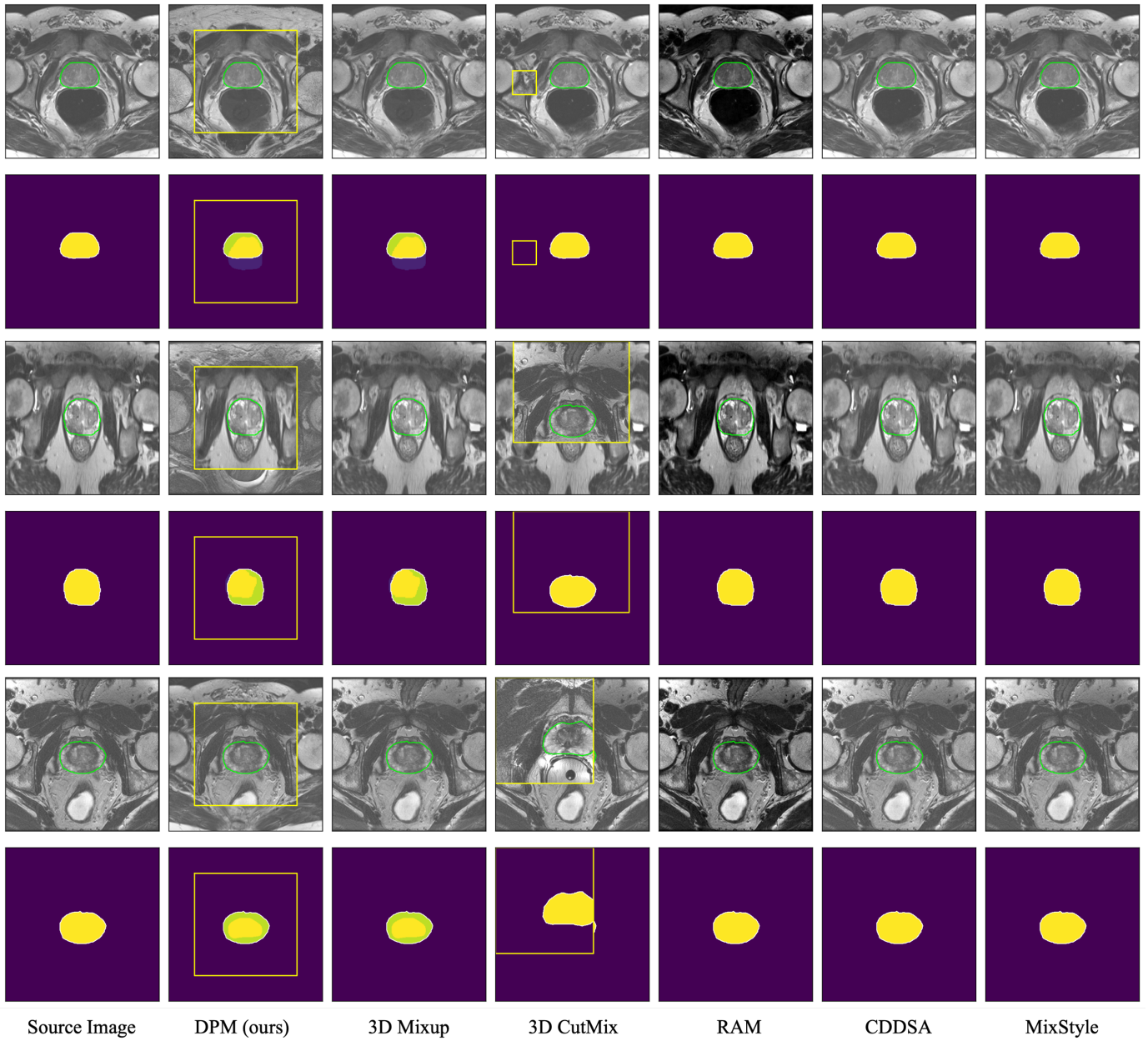


Fig. 1. More visual examples of different augmentation methods on 3D patch.